

With technology threats getting more sophisticated than ever, network intrusion prevention is one of the most challenging tasks for today's IT admins, who have to not only play a defensive game but also discover and understand the innovative threats coming their way. **Diksha P Gupta** from **Open Source For You** spoke to experts from **Juniper Networks**, **Cisco** and **McAfee India** to understand their views on how IT admins should be prepared to face the threats resulting from the current BYOD (bring your own device) trend and the cloud.

The cloud phenomenon is yet another challenge faced by the IT managers as far as network security is concerned. Please share some tips on how IT admins ought to deal with this situation.

There are applications that ensure such protection these days. The IT admins need to ensure that they have control over whatever devices they are dealing with, in terms of either pushing a patch to the device users or even pushing them out of the network. Depending upon the continuity of the application, we can have a client (a Web-based client or a downloadable application), which can provide security against viruses in your device, and will ensure that your information is secured. IT admins have the full control of the device this way and know that their network is not compromised.

IT admins have to face the challenge of building an infrastructure, which is conducive, self-defending and intelligent enough to keep the company's information and intellectual property secure. The cloud is a challenging and unavoidable concept. Though the applications are moving to a non-controlled environment, there are ways of ensuring security in them as well. IT admins should identify what data they are sending on the cloud, what kind of access they can offer and to whom.

The cloud is an interesting concept but needs to be handled with care. What applications are being moved to the cloud and how they are accessed makes a lot of difference. A clear idea on these two things can make the lives of IT admins easier. Resorting to a self-protecting cloud environment is the first step that an IT admin can take to ensure security.

What, according to you, constitutes a network readiness checklist for IT admins?

Network readiness, according to me, would be when, before you roll out a network, you understand whom this network serves. Barring the telecom sector, a network is a service that is being given to a business and doesn't yield any financial results, but is a support function. In that case, you must understand what business is being supported and what its key requirements are. Assessing the risk factors involved in the business is also an important function of an IT admin. One can draw up a good network plan only on the basis of these evaluations.

The checklist is rather simple and involves just three steps. The first one is defence, which means any network should always be defended against external threats. The second important step is about discovery. The modern day threats are very smart and there are greater possibilities of breaches happening, but one can only know about a breach when one discovers it. So discovering a breach is an important aspect to curing it. Discovery in this context is the detection of threats in the network and network behaviour analysis. The last step is the mitigation of the threat. Earlier, it was more or less done manually but now this is an automated process.

One of the most important things in the world of network security is a firewall. Although they have been around for a while, upgrading them to next-gen technology is what every enterprise needs. It offers access control and a higher level of safety. Beyond firewalls, one also needs network intrusion prevention solutions and protection against advanced malware. The challenge with targeted attacks is that an IT admin cannot keep waiting for it to happen and then go for a solution. You cannot protect your network against zero-day, so being well prepared beforehand is a better idea.